

```

240
241 • with years_sales as (
242     select year(order_date) as Years, round(sum(sales),2) as Total_sales from superstore
243     group by year(Order_Date)
244     order by year(Order_Date)
245 )
246     select years,
247     Total_sales,
248     sum(Total_sales) over(order by years)
249     as cumulative_sales
250     from years_sales
251     order by years;

```

Result Grid |  Filter Rows: | Export:  | Wrap Cell Content: 

	years	Total_sales	cummulative_sales
▶	2014	481763.8	481763.8
	2015	464426.24	946190.04
	2016	601265.26	1547455.3
	2017	724994.56	2272449.8600000003

```

228
229 • with year_sales as(
230     select year(order_date ) as Years,
231     round(sum(sales),2) as Total_sales
232     from superstore
233     group by year(order_date)
234     order by year(Order_Date) asc
235 )
236     select years,Total_sales,
237     lag(Total_sales) over(order by years) as Previous_year_sales,
238     round(total_sales - lag(Total_sales) over(order by years),2) as sales_diff
239     from year_sales;
240

```

Result Grid |  Filter Rows: | Export:  | Wrap Cell Content: 

	years	Total_sales	Previous_year_sales	sales_diff
▶	2014	481763.8	NULL	NULL
	2015	464426.24	481763.8	-17337.56
	2016	601265.26	464426.24	136839.02
	2017	724994.56	601265.26	123729.3

```

213
214 • with products as (
215     select category,product_name, round(sum(sales),2) as Total_sales from superstore
216     group by category,product_name
217 ),
218 ranked_product as (
219     select category,product_name,Total_sales,
220     rank() over(partition by category order by total_sales desc)
221     as sales_rank
222     from products
223 )
224 select *
225 from ranked_product
226 where sales_rank <=3;
227
228

```

Result Grid |  Filter Rows: | Export:  | Wrap Cell Content: 

	category	product_name	Total_sales	sales_rank
►	Furniture	HON 5400 Series Task Chairs for Big and Tall	21870.58	1
	Furniture	Riverside Palais Royal Lawyers Bookcase, Royal...	15610.97	2
	Furniture	Bretford Rectangular Conference Table Tops	12995.29	3
	Office Supplies	Fellowes PB500 Electric Punch Plastic Comb Bind...	27453.38	1
	Office Supplies	GBC DocuBind TL300 Electric Binding System	19823.48	2
	Office Supplies	GBC Ibimaster 500 Manual ProClick Binding System	19024.5	3
	Technology	Canon imageCLASS 2200 Advanced Copier	61599.82	1
	Technology	Cisco TelePresence System EX90 Videoconferen...	22638.48	2
	Technology	Hewlett Packard LaserJet 3310 Copier	18839.69	3

```

203
204 • with product_sales as (
205     select category,product_name,
206     round(sum(sales),2) as Total_sales
207     from superstore
208     group by Category,product_name
209 )
210 select category,product_name,Total_sales,
211 rank() over(partition by category order by total_sales desc) as sales_rank
212 from product_sales;

```

Result Grid | Filter Rows: | **Export:** | **Wrap Cell Content:** | **Fetch rows:**

	category	product_name	Total_sales	sales_rank
►	Furniture	HON 5400 Series Task Chairs for Big and Tall	21870.58	1
	Furniture	Riverside Palais Royal Lawyers Bookcase, Royal...	15610.97	2
	Furniture	Bretford Rectangular Conference Table Tops	12995.29	3
	Furniture	Global Troy Executive Leather Low-Back Tilter	12975.38	4
	Furniture	SAFCO Arco Folding Chair	11572.78	5
	Furniture	DMI Eclipse Executive Suite Bookcases	11046.61	6
	Furniture	Hon Deluxe Fabric Upholstered Stacking Chairs,...	10637.53	7
	Furniture	Chromcraft Bull-Nose Wood Oval Conference T...	9917.64	8
	Furniture	Bush Advantage Collection Racetrack Conferen...	9544.72	9
	Furniture	GuestStacker Chair with Chrome Finish Legs	9070.94	10
	Furniture	Office Star - Professional Matrix Back Chair with...	8774.5	11

Result 71 x

```

181 with product_summary as (
182     select product_name,
183     round(sum(sales),2) as Total_sales,
184     round(sum(profit),2) as Total_profit
185     from superstore
186     group by product_name
187 ),
188 product_average as(
189     select avg(Total_sales) as avg_sales
190     from product_summary
191 )
192 select
193 p.product_name,
194 round(p.Total_sales,2) as Total_sales,
195 round(p.Total_profit,2) as Total_profit
196 from product_summary p
197 cross join product_average a
198 where p.Total_sales > a.avg_sales
199 and p.Total_profit <0
200 order by p.Total_sales ;
201

```

Result Grid



Filter Rows:

Export:



Wrap Cell Content:



	product_name	Total_sales	Total_profit
▶	Bush Somerset Collection Bookcase	1263.96	-56.32
	Tennsco Industrial Shelving	1281.44	-108.58
	Hon 30" x 60" Table with Locking Drawer"	1296.03	-177.35
	Eldon Cleatmat Plus Chair Mats for High Pile Car...	1304.13	-111.33
	Office Star - Ergonomically Designed Knee Chair	1311.88	-42.11
	Novimex Turbo Task Chair	1341.52	-32.65
	Safco Industrial Wire Shelving	1363.06	-126.71
	Sauder Camden County Collection Library	1368.26	-103.48
	Cisco CP-7937G Unified IP Conference Station P...	1391.4	-55.66
	Global Armless Task Chair, Royal Blue	1396.44	-20.73
	Fellowes Bankers Box Stor/Drawer Steel Plus	1426.31	-12.79

```

152
153 with customer_summary as(
154     select customer_name,
155     round(sum(sales)) as Total_sales ,
156     round(sum(profit)) as total_profit
157     from superstore
158     group by Customer_name
159 ),
160 customer_average as(
161     select
162     avg(Total_sales) as avg_sales,
163     avg(Total_profit) as avg_profit
164     from customer_summary
165 )
166 select
167     c.customer_name,
168     round(Total_sales,2) as Total_sales,
169     round(Total_profit,2) as Total_profit,
170     CASE
171         WHEN c.total_sales > a.avg_sales
172             AND c.total_profit > a.avg_profit
173         THEN 'High-Value'

```

Result Grid



Filter Rows:

Export:



Wrap Cell Content:

	customer_name	Total_sales	Total_profit	customer_segment
▶	Sean Miller	25043	-1981	Regular
	Tamara Chand	19018	8964	High-Value
	Raymond Buch	15117	6976	High-Value
	Tom Ashbrook	14596	4704	High-Value
	Adrian Barton	14356	5439	High-Value
	Sanjit Chand	14142	5757	High-Value
	Ken Lonsdale	14072	769	High-Value
	Hunter Lopez	12873	5622	High-Value
	Sanjit Engle	12209	2651	High-Value
	Christopher Conant	12129	2177	High-Value
	Todd Sumrall	11886	2369	High-Value

```
138
139 • select category,sum(sales) as Total_sales
140 from superstore
141 group by category
142 having sum(profit) > (
143 select avg(Total_profit)
144 from(
145 select category,sum(profit) as Total_profit
146 from superstore
147 group by category
148 ) as avg_profit );
149
```

Result Grid



Filter Rows:

Export:



Wrap Cell Co

	category	Total_sales
►	Office Supplies	703502.9280000031
	Technology	835900.0669999964


```

127 • SELECT customer_name,
128       ROUND(SUM(sales)) AS Total_sales
129 FROM superstore
130 GROUP BY customer_name
131 HAVING ROUND(SUM(sales)) >(
132 select avg(Total_sales) from(
133 select customer_name, round(sum(sales)) as Total_sales
134 from superstore
135 group by customer_name
136 ) avg_sales
137 );
138

```

Result Grid |   Filter Rows: | Export:  | Wrap Cell Content: 

	customer_name	Total_sales
▶	Brosina Hoffman	6255
	Irene Maddox	4930
	Pete Kriz	8159
	Tracy Blumstein	4731
	Matt Abelman	4281
	Steve Nguyen	3128
	Linda Cazamias	4481
	Ruben Ausman	3832
	Erin Smith	4649
	Kunst Miller	4909
	Karen Daniels	8282
	Joel Eaton	6400
	Stewart Carmic...	4493
	Christopher Schild	3652
	Gary Mitchum	3079
	Rick Benslev	4557

120

121 • SELECT order_id, customer_name, sales

122 FROM superstore

123 WHERE sales > (SELECT ROUND(AVG(sales), 2) AS avg_sales

124 FROM superstore);

125

Result Grid



Filter Rows:

Export:



Wrap Cell Content:



Fetch rows:



	order_id	customer_name	sales
▶	CA-2016-152156	Claire Gute	261.96
	CA-2016-152156	Claire Gute	731.94
	US-2015-108966	Sean O'Donnell	957.5775
	CA-2014-115812	Brosina Hoffman	907.152
	CA-2014-115812	Brosina Hoffman	1706.184
	CA-2014-115812	Brosina Hoffman	911.424
	CA-2016-161389	Irene Maddox	407.976
	CA-2014-105893	Pete Kriz	665.88
	CA-2015-106320	Emily Burns	1044.63
	US-2015-150630	Tracy Blumstein	3083.43
	CA-2016-117590	Gene Hale	1097.544
	CA-2015-117415	Steve Nguyen	532.3992
	CA-2015-117415	Steve Nguyen	371.168
	CA-2016-105816	Janet Molinari	1029.95
	CA-2016-111682	Ted Butterfield	319.41
	CA-2014-106376	Brendan Sweed	1113.024

```
116 • select order_id, datediff(ship_date,order_date) as date_diff from superstore
117 where datediff(ship_date,order_date) > 5;
118
```

Result Grid



Filter Rows:

Export:



Wrap Cell Content:



Fetch rows:



	order_id	date_diff
▶	US-2015-108966	7
	US-2015-108966	7
	CA-2014-105893	7
	CA-2016-105816	6
	CA-2016-105816	6
	CA-2015-135545	6
	CA-2015-135545	6
	CA-2015-135545	6
	CA-2015-135545	6
	US-2015-134026	6
	US-2015-134026	6
	US-2015-134026	6
	CA-2017-155558	7
	CA-2017-155558	7
	US-2017-109484	6
	CA-2016-128867	7

113

114 • `select round(avg(datediff(ship_date,order_date)),2) as avg_date_diff from superstore;`

115

Result Grid |   Filter Rows: | Export:  | Wrap Cell Content: 

	avg_date_diff
▶	3.95

```
108
109 • select month(order_date) as months, round(sum(sales),2) as Total_sales from superstore
110 group by months
111 order by total_sales desc
112 limit 1;
113
```

Result Grid |   Filter Rows: | Export:  | Wrap Cell Content:  | Fetch rows: 

	months	Total_sales
▶	11	348834.56

```
104 -- Date Functions
105 • select year(order_date) as years, round(sum(sales),2) as total_sales, round(sum(profit),2) as Total_profit from superstore
106 group by years
107 order by years,total_sales desc;
108
```

Result Grid |   Filter Rows: | Export:  | Wrap Cell Content: 

	years	total_sales	Total_profit
►	2014	481763.8	49044.43
	2015	464426.24	60907.69
	2016	601265.26	80130.65
	2017	724994.56	92774.99

```

93 • select
94   case
95     when discount < 0.10 then "Low_profit"
96     when discount < 0.30 then "medium_profit"
97     else "High_profit"
98   end discount_category,
99   SUM(Profit) AS total_profit
100 from superstore
101 GROUP BY discount_category
102 ORDER BY total_profit DESC;
103

```

Result Grid



Filter Rows:

Export:



Wrap Cell Co

	discount_category	total_profit
▶	Low_profit	317184.04429999966
	medium_profit	99827.47270000004
	High_profit	-134153.76280000014

```

86 • select order_id,profit,
87     case
88     when profit >0 then "profitable"
89     when profit <0 then "loss_making"
90     end as profit_behaviour
91     from superstore;
92

```

Result Grid



Filter Rows:

Export:



Wrap

	order_id	profit	profit_behaviour
▶	CA-2016-152156	41.9136	profitable
	CA-2016-152156	219.582	profitable
	CA-2016-138688	6.8714	profitable
	US-2015-108966	-383.031	loss_making
	US-2015-108966	2.5164	profitable
	CA-2014-115812	14.1694	profitable
	CA-2014-115812	1.9656	profitable
	CA-2014-115812	90.7152	profitable
	CA-2014-115812	5.7825	profitable
	CA-2014-115812	34.47	profitable
	CA-2014-115812	85.3092	profitable
	CA-2014-115812	68.3568	profitable
	CA-2017-114412	5.4432	profitable
	CA-2016-161389	132.5922	profitable
	US-2015-118983	-123.858	loss_making
	US-2015-118983	-3.816	loss_making


```

76      -- CASE WHEN
77      • select order_id,customer_name,sales,
78      case
79      when sales >15000 then "high_sales"
80      when sales >8000 then "medium_sales"
81      else "low_sales"
82      end as sales_parameter
83      from superstore;
84

```

Result Grid   Filter Rows: | Export:  | Wrap Cell Content: 

	order_id	customer_name	sales	sales_parameter
▶	CA-2016-152156	Claire Gute	261.96	low_sales
	CA-2016-152156	Claire Gute	731.94	low_sales
	CA-2016-138688	Darrin Van Huff	14.62	low_sales
	US-2015-108966	Sean O'Donnell	957.5775	low_sales
	US-2015-108966	Sean O'Donnell	22.368	low_sales
	CA-2014-115812	Brosina Hoffman	48.86	low_sales
	CA-2014-115812	Brosina Hoffman	7.28	low_sales
	CA-2014-115812	Brosina Hoffman	907.152	low_sales
	CA-2014-115812	Brosina Hoffman	18.504	low_sales
	CA-2014-115812	Brosina Hoffman	114.9	low_sales
	CA-2014-115812	Brosina Hoffman	1706.184	low_sales
	CA-2014-115812	Brosina Hoffman	911.424	low_sales
	CA-2017-114412	Andrew Allen	15.552	low_sales
	CA-2016-161389	Irene Maddox	407.976	low_sales
	US-2015-118983	Harold Pawlan	68.81	low_sales
	US-2015-118983	Harold Pawlan	2.544	low_sales

71

72 • `select sub_category, round(sum(profit),2) as total_profit from superstore`

73 `where profit <0`

74 `group by sub_category;`

75

Result Grid



Filter Rows:

Export:



Wrap Cell Content:



	sub_category	total_profit
►	Tables	-32412.15
	Appliances	-8629.64
	Binders	-37676.07
	Chairs	-9880.84
	Bookcases	-12152.21
	Furnishings	-6043.72
	Storage	-6087.79
	Accessories	-930.63
	Phones	-7530.62
	Machines	-30118.67
	Supplies	-3015.62
	Fasteners	-33.2

66

```
67 • select region, round(sum(sales),2) as Total_sales from superstore  
68 group by region  
69 order by total_sales desc;
```

70

Result Grid



Filter Rows:

Export:



Wrap Cell Content:



	region	Total_sales
▶	West	713471.34
	East	672194.05
	Central	497800.87
	South	388983.59

```

63 • select customer_name, round(sum(sales),2) as total_sales from superstore
64 where sales >5000
65 group by Customer_name;
66

```

Result Grid



Filter Rows:

Export:



Wrap Cell Content:



	customer_name	total_sales
▶	Becky Martin	8159.95
	Christopher Martinez	6354.95
	Grant Thornton	7999.98
	Ken Lonsdale	8187.65
	Tom Ashbrook	11199.97
	Sean Miller	22638.48
	Sanjit Chand	9449.95
	Hunter Lopez	10499.97
	Bill Shonely	9099.93
	Todd Sumrall	5199.96
	Christopher Conant	8399.98
	Andy Reiter	5443.96
	Tom Boeckenhauer	6999.96
	Tamara Chand	17499.95
	Daniel Raglin	5399.91
	Ravmond Buch	13999.96

Result Editor

```

54 • SELECT
55     sub_category,
56     MAX(profit) AS max_profit,
57     MIN(profit) AS min_profit
58 FROM
59     superstore
60 GROUP BY sub_category
61 ORDER BY max_profit desc;
62

```

Result Grid



Filter Rows:

Export:



Wrap

	sub_category	max_profit	min_profit
▶	Copiers	8399.976	59.998
	Binders	4946.37	-3701.8928
	Machines	2799.984	-6599.978
	Phones	1228.1787	-386.3916
	Bookcases	1013.127	-1665.0522
	Accessories	829.3754	-75.5958
	Appliances	793.716	-1181.2824
	Storage	792.2691	-337.806
	Chairs	770.352	-630.882
	Tables	629.01	-1862.3124
	Furnishings	387.5676	-427.45
	Labels	385.3752	0.6786
	Paper	352.296	1.07
	Supplies	327.506	-1049.3406
	Envelopes	204.0714	0.5508
	Art	112.574	0.1533

```

45 • SELECT
46     category,
47     ROUND(SUM(sales), 2) AS total_sales,
48     ROUND(SUM(profit), 2) AS Total_profit
49 FROM
50     superstore
51 GROUP BY category
52 ORDER BY total_sales desc;
53

```

Result Grid



Filter Rows:

Export:



Wrap Cell Content:

	category	total_sales	Total_profit
►	Technology	835900.07	145387.1
	Furniture	733046.86	16980.77
	Office Supplies	703502.93	120489.89

```
40 • select max(profit) as highest_profit,  
41       min(profit) as lowest_profit  
42       from superstore;  
43
```

Result Grid |  Filter Rows: | Export:  | Wra

	highest_profit	lowest_profit
▶	8399.976	-6599.978

```
38 • select count(distinct customer_name) as unique_customer from superstore;
39
```

Result Grid



Filter Rows:

Export:



Wrap Cell Content:



	unique_customer
--	-----------------

▶	793
---	-----


```
--  
33 • SELECT  
34     round(SUM(sales) / COUNT(DISTINCT order_id),2) AS avg_order_value  
35 FROM  
36     superstore;
```

Result Grid



Filter Rows:

Export:



Wrap Cell Content:



	avg_order_value
▶	460.85

31 • `select round(avg(discount),2) as avg_discount from superstore;`

Result Grid



Filter Rows:

Export:



Wrap Cell Content:



	avg_discount
▶	0.16

29 •

```
select round(sum(quantity),2) as total_quantity from superstore;
```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	total_quantity
▶	36749

27 • `select round(sum(profit),2) as total_profit from superstore;`

28

Result Grid



Filter Rows:

Export:



Wrap Cell Content:



	total_profit
▶	282857.75

```
24      -- Aggregate Functions
25 •    select round(sum(sales),2) as total_sales from superstore;
26
```

Result Grid



Filter Rows:

Export:



Wrap Cell Content:



	total_sales
▶	2272449.86



```

19 • select category,product_name,region,sales from superstore
20 where region="West"
21 order by sales desc
22 limit 10;
23
24 -- Aggregate Functions

```

Result Grid



Filter Rows:

Export:



Wrap Cell Content:



Fetch

	category	product_name	region	sales
►	Technology	Canon imageCLASS 2200 Advanced Copier	West	13999.96
	Office Supplies	High Speed Automatic Electric Letter Opener	West	8187.65
	Office Supplies	Fellowes PB500 Electric Punch Plastic Comb Bind...	West	5083.96
	Office Supplies	High Speed Automatic Electric Letter Opener	West	4912.59
	Technology	Cubify CubeX 3D Printer Double Head Print	West	4799.984
	Office Supplies	Ibico EPK-21 Electric Binding System	West	4535.976
	Technology	Okidata MB760 Printer	West	4476.8
	Technology	Apple iPhone 5	West	4158.912
	Furniture	Bretford Rectangular Conference Table Tops	West	3610.848
	Furniture	DMI Eclipse Executive Suite Bookcases	West	3406.664

```

14 • SELECT category,product_name,discount
15 FROM superstore
16 WHERE discount >= 0.30
17 limit 10;
18
19 • select * from superstore

```

Result Grid



Filter Rows:

Export:



Wrap Cell Conte

	category	product_name	discount
►	Furniture	Bretford CR4500 Series Slim Rectangular Table	0.45
	Office Supplies	Holmes Replacement Filter for HEPA Air Cleaner...	0.8
	Office Supplies	Storex DuraTech Recyded Plastic Frosted Binders	0.8
	Furniture	Global Deluxe Stacking Chair, Gray	0.3
	Furniture	Riverside Palais Royal Lawyers Bookcase, Royal...	0.5
	Office Supplies	Avery Recyded Flexi-View Covers for Binding S...	0.7
	Furniture	Electrix Architect's Clamp-On Swing Arm Lamp, ...	0.6
	Furniture	Atlantic Metals Mobile 3-Shelf Bookcases, Custo...	0.32
	Furniture	Global Fabric Manager's Chair, Dark Gray	0.3
	Furniture	Global Value Mid-Back Manager's Chair, Gray	0.3

```

10 • select category,product_name,profit from superstore
11 where profit <0
12 limit 10;

```

Result Grid |   Filter Rows: | Export:  | Wrap Cell Content:

category	product_name	profit
Furniture	Bretford CR4500 Series Slim Rectangular Table	-383.031
Office Supplies	Holmes Replacement Filter for HEPA Air Cleaner...	-123.858
Office Supplies	Storex DuraTech Recycled Plastic Frosted Binders	-3.816
Furniture	Global Deluxe Stacking Chair, Gray	-1.0196
Furniture	Riverside Palais Royal Lawyers Bookcase, Royal...	-1665.0522
Office Supplies	Avery Recycled Flexi-View Covers for Binding S...	-7.0532
Furniture	Electrix Architect's Clamp-On Swing Arm Lamp, ...	-147.963
Furniture	Atlantic Metals Mobile 3-Shelf Bookcases, Custo...	-46.9764
Furniture	Global Fabric Manager's Chair, Dark Gray	-15.147
Furniture	Global Value Mid-Back Manager's Chair, Gray	-15.2225


```

5      -- SELECT, WHERE, ORDER BY
6      • select category,product_name,sales from superstore
7      order by sales desc
8      limit 10;

```

Result Grid   Filter Rows: Export:   Wrap Cell Cont			
	category	product_name	sales
►	Technology	Cisco TelePresence System EX90 Videoconferen...	22638.48
	Technology	Canon imageCLASS 2200 Advanced Copier	17499.95
	Technology	Canon imageCLASS 2200 Advanced Copier	13999.96
	Technology	Canon imageCLASS 2200 Advanced Copier	11199.968
	Technology	Canon imageCLASS 2200 Advanced Copier	10499.97
	Office Supplies	GBC Ibimaster 500 Manual ProClick Binding System	9892.74
	Office Supplies	Ibico EPK-21 Electric Binding System	9449.95
	Technology	3D Systems Cube Printer, 2nd Generation, Mag...	9099.93
	Technology	HP Designjet T520 Inkjet Large Format Printer -...	8749.95
	Technology	Canon imageCLASS 2200 Advanced Copier	8399.976